

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

001 This supplement records the material that supports
002 the main paper without being required to follow its
003 argument: the full metric specification, the complete
004 validation matrix, the evaluation-set composition, per-
005 horizon results beyond the two principal horizons, and
006 the exploratory image-conditioned analysis. Section
007 numbering is independent of the main paper; cross-
008 references in the main text point here by name.

009 1. Region Partition

010 Derivation. For every generated sequence we average
011 optical-flow magnitude over the leading 12% of sam-
012 pled frame pairs and threshold it by Otsu’s method
013 into a dynamic region and its complement. The static
014 region is then reduced by a fixed morphological ero-
015 sion, and a 4% frame border is discarded; together
016 these form the ignored transition band, so boundary
017 pixels—where the partition is least certain and flow
018 least reliable—contribute to neither region. The same
019 procedure is applied to both tracks.

020 Why the estimate is restricted to an early window.
021 Deriving a partition from a sequence’s entire motion
022 field would be circular, since accumulated background
023 drift would itself be read as evidence that the affected
024 region is dynamic. Restricting the estimate to the early
025 window breaks that circularity in the direction that
026 matters: at the 60s horizon the partition is fixed from
027 the first 7.1s, before long-horizon drift has accumu-
028 lated, and the metrics are then evaluated over the re-
029 mainder.

030 Limitations. The partition is obtained automatically
031 rather than by annotation, so no inter-annotator agree-
032 ment is reported, and for the image-conditioned track
033 it is derived per generated sequence rather than once
034 from the shared source image. The early window is
035 a weaker guarantee than a first-frame annotation: a
036 system that drifts within the first 12% of its rollout
037 can influence its own partition. We therefore treat
038 partitions as measured quantities and report the con-
039 sequence directly—every measurement is recomputed
040 under systematic erosion and dilation of the boundary,
041 with the resulting response given in Table 1. A human-
042 verified, first-frame annotation protocol is the natural
043 strengthening of this component.

perturbation	clips	fBD	NBF	MCFF-L	FP	DLR	DAR	VB-DD
photometric	12	-0.30	+0.90	+1.00	+1.00	-1.00	-0.90	+1.00
mask radius	12	+0.70	+1.00	+0.90	-0.70	+1.00	-0.70	—
translation	12	+1.00	+1.00	-1.00	+0.60	+1.00	+0.90	+0.70
rotation	12	+0.70	+0.60	-0.70	+0.30	-0.80	+1.00	+0.70
scale	12	+1.00	+1.00	-0.90	-0.60	+1.00	+1.00	+1.00
attenuation	12	-0.40	-1.00	+0.00	+0.40	+1.00	-1.00	-1.00
freeze	12	-1.00	-1.00	-1.00	-0.80	-0.20	-0.80	-1.00
repetition	12	-0.20	-0.20	+0.20	+0.20	-0.40	-0.40	-0.20

Table 1. Mechanistic validation. Spearman correlation be-
tween injected severity and each factor’s response, over con-
trolled perturbations of real fixed-camera clips. Signs are
fixed in advance by each definition. mask radius perturbs
the region partition rather than the video, so near-zero en-
tries indicate the desired insensitivity. Cells where a factor
responds weakly or non-monotonically are reported rather
than omitted.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth	-0.96	-0.82	-0.61	-0.64
VB-flick	-0.96	-0.82	-0.61	-0.64

Table 2. Rank disagreement at 60s. Spearman correla-
tion between generic whole-frame metrics and SNF-Bench
factors over method-level means. A positive correlation be-
tween apparent motion and background flow is the effect
the benchmark isolates.

044 2. Complete Validation Matrix

045 Geometric severity is specified in a common physical
046 unit—the mean pixel displacement a corruption in-
047 duces over the frame by the end of the rollout—and
048 each family’s own parameter is derived from it using
049 the frame geometry: a translation of d pixels, a rota-
050 tion of arctan-free angle $\theta = d/\bar{r}$ radians, or a zoom of
051 $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame
052 centre. Expressing severity this way is what makes the
053 geometric families comparable; a parameter expressed
054 in degrees is not commensurate with one expressed in
055 pixels, and treating them as though they were under-
056 states angular corruptions by more than an order of
057 magnitude at this resolution.

category	definition	240s tier is exploratory; both are given here for com-	069
river_stream	channel water: rivers, streams, creeks, plateaus, waterfalls, rapids, floods		070
ocean_waves	open water with wave action: surf, tides, coastal swell, seascapes		
precipitation	falling water or ice: rain, snowfall, and the surfaces they wet		
fire_smoke	combustion plumes: wildfire, campfire, house fire, etc.	1. Signed Drift Attenuation	071
lava_volcanic	volcanic flow and ejecta: lava channels, ash columns, eruption smoke		
windborne	air-carried particulates and wind-driven vegetation: dust, sand, blossom, leaves, thrashing trees, drifting cloud	Drift attenuation is stored as a signed quantity and	072
track/duration	n	reported clipped to [0, 1]. Negative values are informa-	073
i2v/5s	10	tive rather than anomalous: where local flow opposes	074
i2v/60s	30	the estimated global field, compensation increases the	075
i2v/120s	20	measured magnitude. We report their incidence rather	076
i2v/240s	5	than suppressing it.	077
t2v/5s	6		
t2v/60s	23		
t2v/120s	6		
t2v/240s	4		
track	source	8. Deployment Sensitivity	078
i2v	meta		
t2v	keyword	9 Interpretation Changes	079
t2v	keyword-body		
t2v	override		
track	duration	Cases are selected by a fixed rank-gap rule applied	080
t2v	60s	to every audited system, so no qualifying case can be	081

Table 3. Scene-category balance of the evaluation set, by track and horizon.

Method	status	setting	5s	10s	240s	240s
CausVid	public	native	V6 B	V23 T23 B	V6 B	V4 B
Self-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Infinite-Forcing	public	native	V6 B	V24 T24 B	V6 B	V4 B
Rolling-Forcing	public	native	V6 B	V24 T24 B	V6 B	V4 B
Reward-Forcing	public	native	V12 B	V23 T23 B	V6 B	V4 B
LongLive	public	native	V6 B	V23 T23 B	V6 B	V4 B
Causal-Forcing	public	native	V6 B	V23 T23 B	V6 B	V4 B
Method	status	setting	5s	10s	240s	240s
Causal-Forcing++ (2-step)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (1-step)	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T1/5 X B
Causal-Forcing (framework)	public	matched	V10 T10 X B	V30 T1/30 X B	V20 T1/20 X B	V5 T5 X B
Causal-Forcing++ (2-step, native nfpb=1)	public	native	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Self-Forcing	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
CausVid	public	matched	V10 T10 X B	V30 T1/30 X B	V20 T1/20 X B	V5 T5 X B
Wan2.1-I2V-14B-480P	public	native	V10 T10 X B	.	.	.
Wan2.2-I2V-A14B	public	native	V10 T10 X B	.	.	.
LTX-Video 13B-0.9.8-distilled	public	native	V10 T10 X B	.	.	.
Causal-Forcing++ (1-step, native)	public	native	.	V30 T30	V20 T20	.

Table 4. Asset and score coverage for the audited public systems. Tn/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

058 3. Whole-Frame and SNF-Bench Rank Corre-
059 lation

060 4. Evaluation-Set Composition

061 5. Asset and Score Coverage

062 Coverage is counted from valid per-video measure-
063 ments rather than from the presence of a results file,
064 so an entry whose measurement run terminated early
065 is reported as partial rather than complete.

066 6. Results at All Horizons

067 The main paper reports the two principal horizons.
068 The 5s tier serves as an initialisation check and the

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 5. Complete T2V audit, all horizons. The main paper reproduces the 60s and 120s panels; the 5s tier is an initialisation check and the 240s tier is diagnostic. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

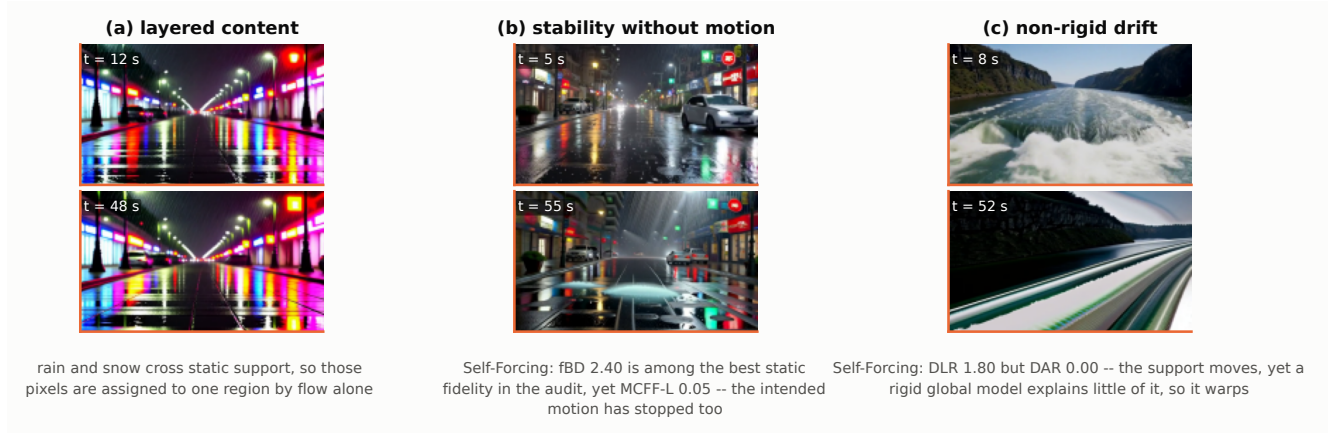


Figure 1. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation identifies support that moves without translating, which a rigid global model does not explain.

Method	track	dur	n	negative	rate
CausVid	i2v	120s	1	1	100.0%
CausVid	i2v	240s	5	0	0.0%
CausVid	i2v	5s	10	6	60.0%
CausVid	i2v	60s	1	0	0.0%
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%
Causal-Forcing++ (1-step)	i2v	240s	1	0	0.0%
Causal-Forcing++ (1-step)	i2v	5s	10	0	0.0%
Causal-Forcing++ (1-step)	i2v	60s	30	0	0.0%
Causal-Forcing (framework)	i2v	120s	1	0	0.0%
Causal-Forcing (framework)	i2v	240s	5	0	0.0%
Causal-Forcing (framework)	i2v	5s	10	2	20.0%
Causal-Forcing (framework)	i2v	60s	1	0	0.0%
Causal-Forcing++ (2-step)	i2v	120s	20	0	0.0%
Causal-Forcing++ (2-step)	i2v	240s	5	0	0.0%
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%
Causal-Forcing++ (2-step)	i2v	60s	30	0	0.0%
Causal-Forcing++ (1-step, native)	i2v	120s	20	1	5.0%
Causal-Forcing++ (1-step, native)	i2v	60s	30	3	10.0%
Causal-Forcing++ (2-step, native nfpb=1)	i2v	120s	20	3	15.0%
Causal-Forcing++ (2-step, native nfpb=1)	i2v	240s	5	2	40.0%
Causal-Forcing++ (2-step, native nfpb=1)	i2v	5s	10	1	10.0%
Causal-Forcing++ (2-step, native nfpb=1)	i2v	60s	30	2	6.7%
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	4	40.0%
Self-Forcing	i2v	120s	20	3	15.0%
Self-Forcing	i2v	240s	5	2	40.0%
Self-Forcing	i2v	5s	10	2	20.0%
Self-Forcing	i2v	60s	30	8	26.7%
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%
CausVid	t2v	120s	6	0	0.0%
CausVid	t2v	240s	4	1	25.0%
CausVid	t2v	5s	6	2	33.3%
CausVid	t2v	60s	23	0	0.0%
Causal-Forcing	t2v	120s	6	0	0.0%
Causal-Forcing	t2v	240s	4	0	0.0%
Causal-Forcing	t2v	5s	6	2	33.3%
Causal-Forcing	t2v	60s	23	3	13.0%
Infinite-Forcing	t2v	120s	6	0	0.0%
Infinite-Forcing	t2v	240s	4	1	25.0%
Infinite-Forcing	t2v	5s	6	0	0.0%
Infinite-Forcing	t2v	60s	23	4	17.4%
LongLive	t2v	120s	6	1	16.7%
LongLive	t2v	240s	4	0	0.0%
LongLive	t2v	5s	6	1	16.7%
LongLive	t2v	60s	23	2	8.7%
Reward-Forcing	t2v	120s	6	1	16.7%
Reward-Forcing	t2v	240s	4	0	0.0%
Reward-Forcing	t2v	5s	12	2	16.7%
Reward-Forcing	t2v	60s	23	3	13.0%
Rolling-Forcing	t2v	120s	6	3	50.0%
Rolling-Forcing	t2v	240s	4	0	0.0%
Rolling-Forcing	t2v	5s	6	1	16.7%
Rolling-Forcing	t2v	60s	24	2	8.3%
Self-Forcing	t2v	120s	6	2	33.3%
Self-Forcing	t2v	240s	4	1	25.0%
Self-Forcing	t2v	5s	6	1	16.7%
Self-Forcing	t2v	60s	23	4	17.4%

Table 6. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow.

Checkpoint	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.18	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.15	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.61	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.06	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.41	-0.14

Table 7. Deployment sensitivity. Change from each checkpoint's native configuration to the common long-horizon wrapper, i.e. the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method's performance.

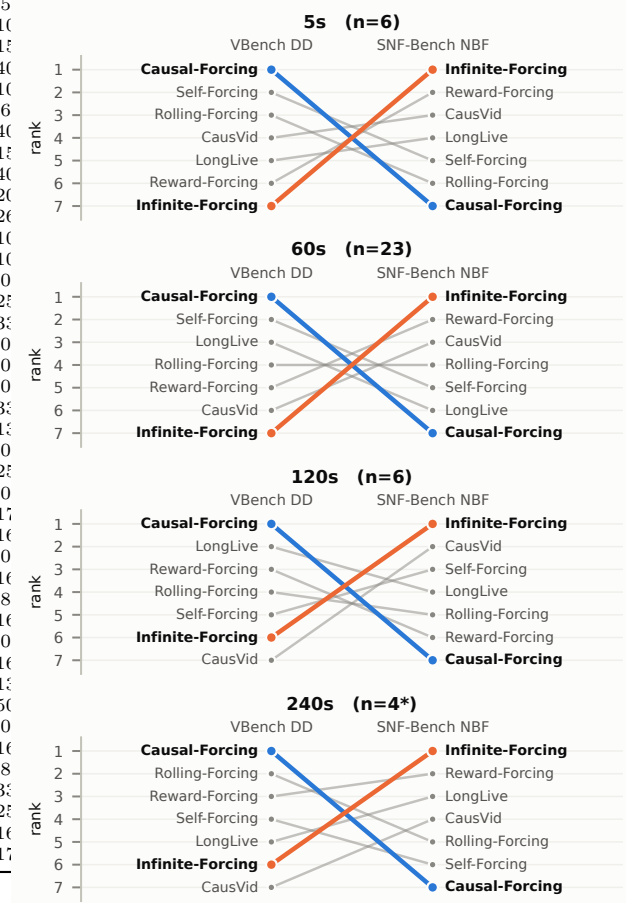


Figure 2. Rank disagreement at every audited horizon. Rank under a whole-frame motion score against rank under normalised background flow. Both quantities are independent of the global-motion estimator. Prompt counts are given per panel; the shorter and longer tiers are diagnostic.

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 8. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	–	0.67	0.86	0.67	0.15
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (framewise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, native nfb=1)	native	9.94	6.91	–	0.81	0.98	0.75	0.11
Self-Forcing	matched	17.3	5.52	–	0.74	0.88	0.67	0.14
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, native)	native	10.4	5.25	–	0.52	0.97	0.83	0.14

Table 9. SNF-Bench audit, I2V track. Setting is stated per row: native systems run under their authors’ configuration, matched systems under the common long-horizon wrapper and are read as a deployment stress test rather than as the published method’s performance. n is stated per horizon and is common to every row of that panel.

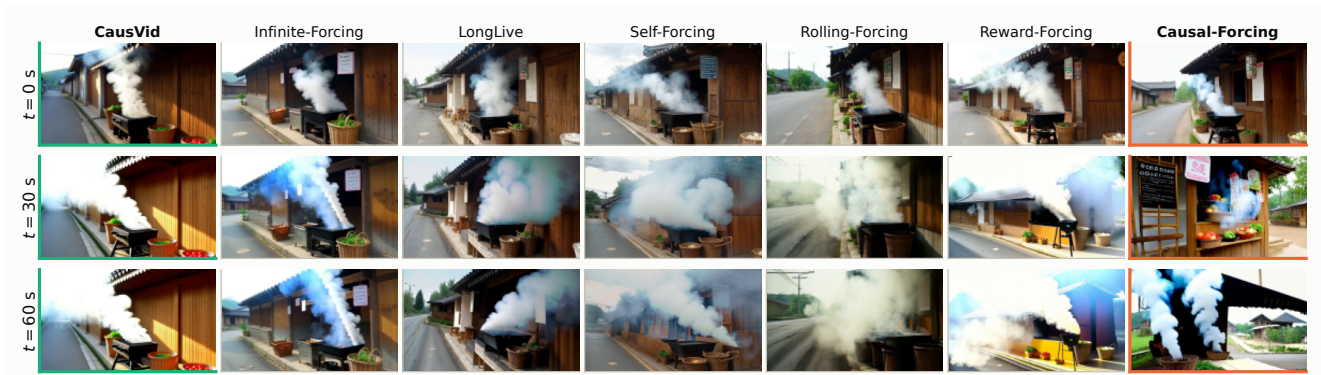


Figure 3. Every audited system on one prompt. Columns are ordered by increasing static-region drift. The two highlighted systems are the pair a whole-frame motion score cannot separate: the leftmost holds its support fixed while its smoke continues to rise, the rightmost has relocated the scene entirely by $t = 30$ s.

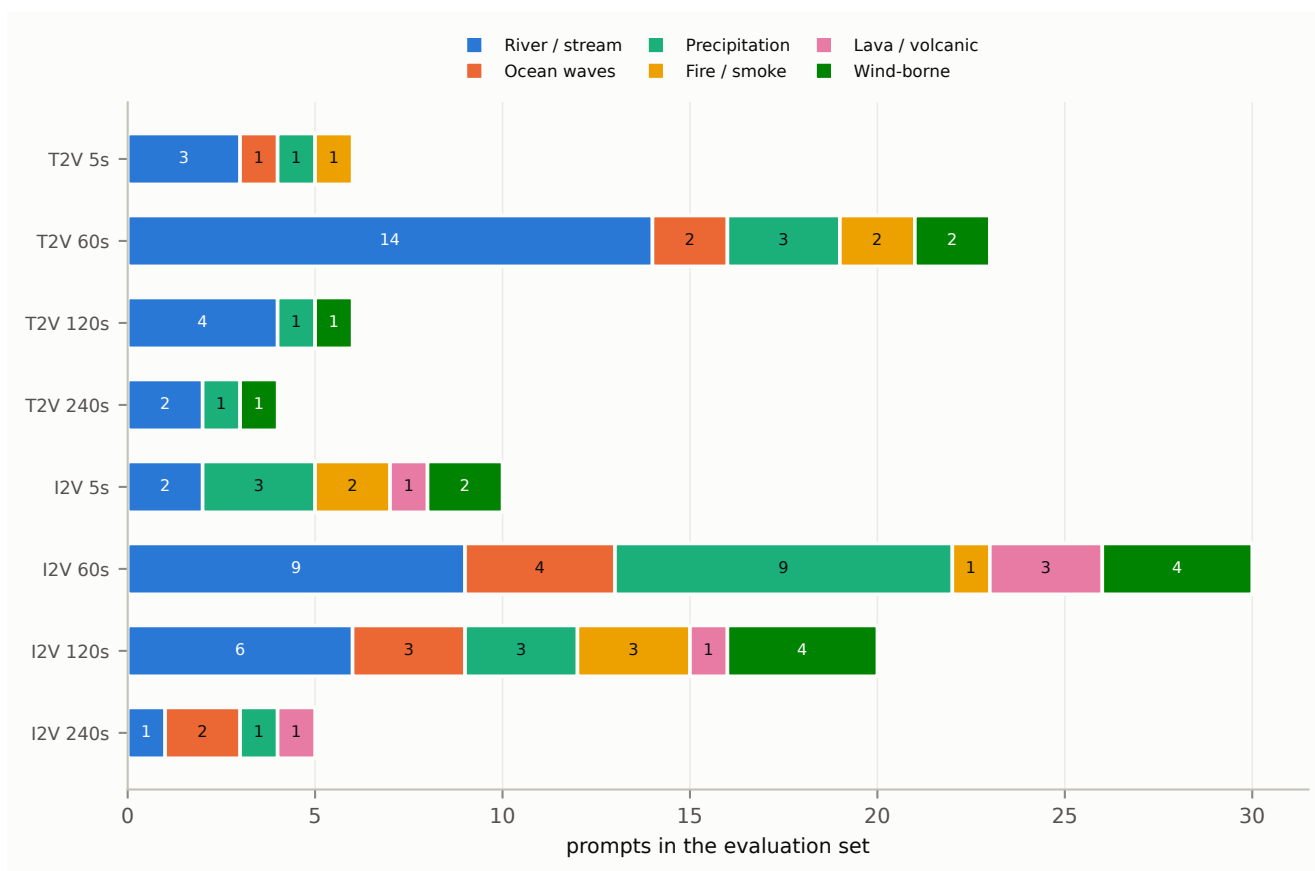


Figure 4. Composition of the evaluation set by scene category. Counts are prompts per category at each track and horizon.